Evaluation & Testing

Scope | the Eval action items from the last two meetings, and where each one is answered

Action item	What we did	Slide
Run round 1 in week 8 (minutes #5)	RAG (11/9) and GraphRAG (11/11) ran our scoring; we reproduced both and fixed what they flagged	2-4
Round 1 end to end (RAG's reader, 11/11)	Reproduced RAG's scores with our graders; fixed the three grading notes in Gold Set v2.2	3
Run one question set on RAG and GraphRAG, side by side	All 84 Microsoft 10-Q questions through each team's own code, same filing, top 8	4
Show where GraphRAG does better, and where RAG does	Plus the 9 Exhibit 21 relationship questions (traversal)	5
Label the routing chart's axes; define the keyword router	Axes titled; bars are coverage, defined, and not accuracy; numbers matched to each method	6
Explain how unneeded calls per question are measured	Definition with a worked example	6
Concrete routing examples; where the router sits	Three real questions walked through; the keyword router is a baseline, and Agents can score theirs with the same script	7
Say accuracy is on the current 155 questions only	Stated on every routing and wording number	3-8
Typos and messy wording, for real-user accuracy	427 rewrites of the 155 questions through gate, router and both retrievers	8

All figures use the current 155 Gold Set v2.2 questions

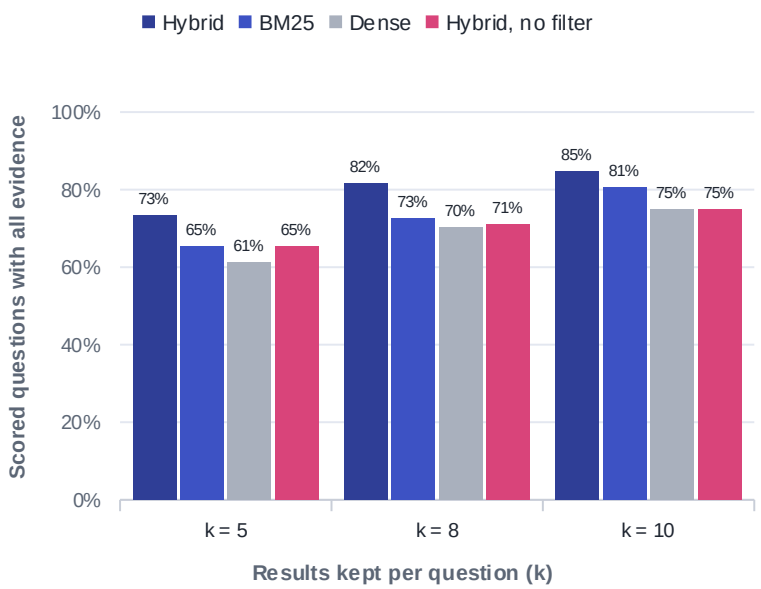
104 finance data, 33 text, 9 relationship, 8 no tool, 1 both. Microsoft has 92 of them, so per-route numbers on 1-9 questions are examples, not estimates. They will move as questions are added. v2.2 keeps v2's questions and answers; it changes grading only (slide 3).

Pinned: RAG main @ 5bfafeb (11/5), GraphRAG main @ 21f0795 (11/11) and Exhibit 21 branch @ c21fe89, Data @ 7040b8a. No LLM is used in Eval's results; the teams' round-1 runs (slides 2-4) are re-scored here from the files they sent.

Source: Week 8/README.md; scripts/run_all.sh (97 tests pass)

Evaluation & Testing

Round 1: retrieval | Gold Set v2, run by RAG with our scoring rules (11/9): 105 of 124 found



83/93 number questions; prose 22/31 (hybrid, k = 10, hit must come from the right filing)

34/37 single-company questions on the nine other companies, vs Microsoft 71/84: adding nine companies did not hurt retrieval

Why 19 were missed	n	Who fixes it
Cross-company ranking: 10-20 figures	3	Agents: one search per company
Our anchors were wrong (Gold v0 conversion)	4	**Eval: fixed in v2.1 (all 22 converted anchors)**
Needs the 10-K and the 10-Q together	3	Agents: one search per form
Fusion pushed it just past k = 10	6	RAG: candidate k, fusion weights
All three methods missed	3	RAG: ranking; 1 is truly multi-part

Also found: a wrong-period figure sits in the top 8 for 90.5% of the 42 number questions that have one, so the reader must be told which column to take; fusion scores look the same for hits and misses, so 'weak evidence' has to come from structure (empty result, no filter match, BM25 and dense disagree).

Source: RAG team, round-1 results 11/9 (KPMG_RAG scripts/run_round1_eval.py), transcribed in Week 8/data/round1_rag_2026-11-09.json; Gold Set v2.1 in Week 8/GOLD_SET_V2.md. Evidence level (retrieval only)

Evaluation & Testing

Round 1: answers | RAG's reader end to end, scored with our graders (11/11)

Measure	RAG's report (subset)	All 155, Gold v2	All 155, Gold v2.2
Numbers right (NAA)	0.935 (87/93)	0.926 (87/94)	**0.947 (89/94)**
Refusal, balanced	0.977	0.944	0.944
Unanswerable refused	10/10	12/12	12/12
Fiscal calendar, lists	n/a	5/12	5/12
False premise corrected	n/a	5/7	6/7
Narrative (28 of 30 answered)	n/a	judge pending	judge pending

Citations, checked in RAG's chunks: all 185 cited chunks exist; every answer cites a filing the gold answer uses (127/127); 58 of 59 lookups show the number in a cited chunk.

Why 16 answerable questions were declined: 7 need Exhibit 21 (GraphRAG's route); 3 need NetApp's 10-Q, which Data holds for review; 3 are cross-company rankings (one search per company); 2 state the answer but are marked not answered (GSv0-MSFT-023, EV1R-005); 1 retrieval miss.

RAG's three grading notes, fixed in Gold Set v2.2

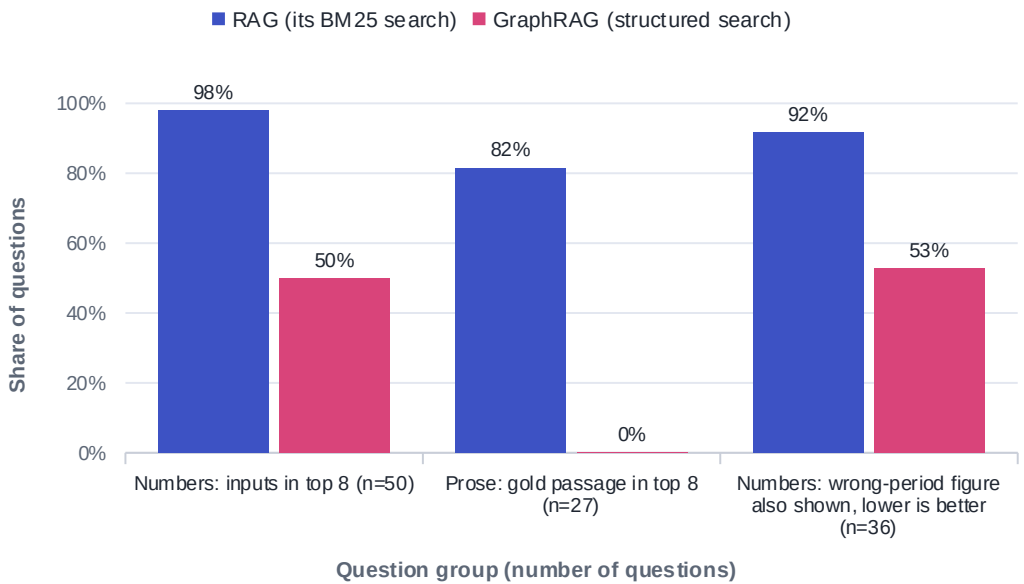
RAG's note	Fix
EV1-015: '$0.91' fails 3 significant figures	Floor = the precision the filing prints: 2 (EV1-027 too)
EV1-043: three figures, one value field	7 multi-part questions need every figure, from an optional values list or the text
EV1R-002, 006: answered from 10-K text	Route text; GraphRAG agreed. Keyword router 84% → 83%
Found from RAG's files	
EV1R-005: 10-K says 25%, 10-Q 27%	Both accepted
v2.1 anchors: 2 sentences in one item	One item per point; all anchors outside Exhibit 21 and NetApp's 10-Q found in RAG's chunks

Next: the narrative judge is ready (point by point, with a quote check). It needs an API key, and it counts only after it agrees with our hand labels.

Source: Week 8/results/answers_rag_reader_round1.json (scripts/grade_answer_blocks.py), results/gold_v2_2_changes.json. RAG's answers and chunks (11/11) stay private

Evaluation & Testing

RAG vs GraphRAG | the same 84 Microsoft 10-Q questions, each team's own code on the same filing



Head to head on the 77 answerable questions

	GraphRAG has it	GraphRAG misses
RAG has it	25	46
RAG misses	0	6

How it was run. Each team's own code: RAG's chunker and Retriever.search; GraphRAG's fact loader on Data's copy of the filing, then structured_search.

GraphRAG's own round 1, all 20 filings (11/11): structured search has the needed number in its top 10 for 63 of 93, at rank 1 for 26; the graph arm, scored for the first time, has the prose anchors for 10 of 38. Our rerun on their store matches all 155 questions.

Neither side returned an empty result on the 7 unanswerable questions, so abstaining stays with the Agents' evidence check.

Read: RAG's search finds the needed numbers and passages more often; GraphRAG's facts carry their exact period, so wrong-period figures show up less. Evidence level only: no reader model.

Source: Week 8/results/side_by_side.json (scripts/run_side_by_side.py). RAG's hybrid with its offline stand-in for the dense model: numbers 82%, prose 85%

Evaluation & Testing

Where each does better | GraphRAG on relationships across filings, RAG on reading the filing itself

9 relationship questions (Exhibit 21)

9/9 answered exactly by traversal of GraphRAG's own Exhibit 21 rows (parent HAS_SUBSIDIARY subsidiary): counts, joins, ownership checks

0/9 reachable by RAG today: it indexes each filing's main report, and Exhibit 21 is a separate document

8/9 with Exhibit 21 added as text, RAG's search brings every needed row; the counting and joining is then left to the reader (Apple and Micron's shared countries: 15 of 31 rows)

Cross-check: GraphRAG's Exhibit 21 pipeline (11/8) and our gold reads agree on 62 of 62 rows, names and jurisdictions. Rows are GraphRAG's; the traversal ran in memory here, since their Neo4j is not reachable from our side.

	Better at	Evidence (Microsoft 10-Q, top 8)
GraphRAG	Relationship questions	Counts and joins come out exact (left)
	Exact periods on numbers	Wrong-period figure shown 53% vs RAG 92%
RAG	Prose and explanations	22/27 vs 0/27: structured search holds numbers only
	Finding the right number	49/50 vs 25/50

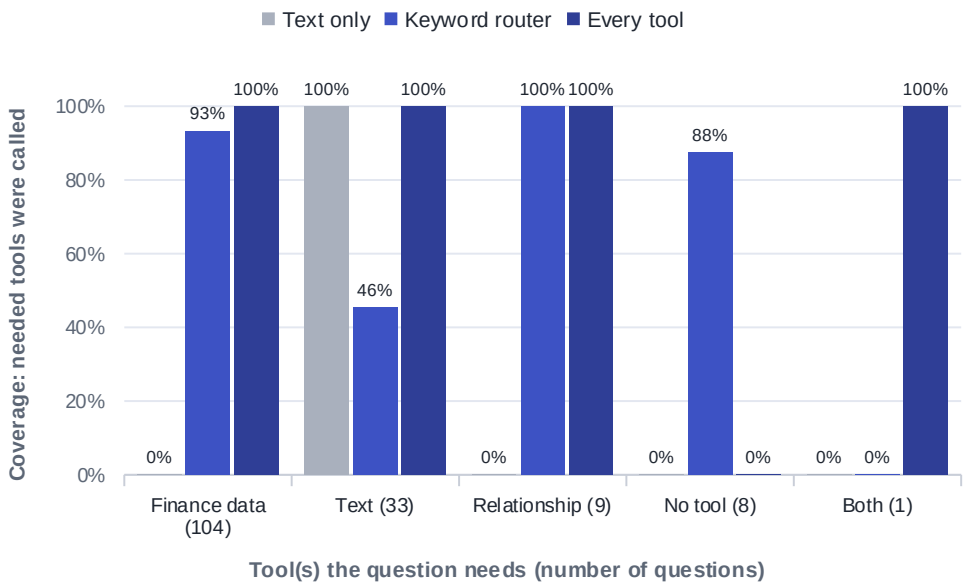
Where GraphRAG's 25 number misses come from (each has a fix)
13 ranked below 8: in 6, 'three months' or 'six months' matches Microsoft 365 ('ThreeSixFive') rows, so total revenue ranks 10th
10 calculations: one call cannot fetch every input; one call per input would
2 wording: 'current unearned revenue', 'repurchasing common stock' miss their concepts
Rank 1: right number 14/50, a wrong-period number 6/36; GraphRAG flagged this itself

Together: route relationship questions to GraphRAG and narrative ones to RAG; numbers can use either once GraphRAG's ranking fix lands, with GraphRAG's period labels as the check.

Source: Week 8/results/side_by_side.json (part_b, number_miss_diagnosis); GraphRAG ex21_pipeline/output/graph_edges.csv @ c21fe89; data/sec_reads/exhibit21_2026-11-05.json; RAG rag/ingest.py

Evaluation & Testing

Routing | coverage: did each method call the tools a question needs? (155 questions)



Coverage = share of questions where every tool the question needs was called (no tool, for the 8 that should be blocked).
It is not accuracy. Calling the right tools does not make the final answer right; answer accuracy needs the reader: RAG's round-1 answers are scored on slide 3.

Method	Coverage	Calls per question	Unneeded calls per question
Text only	21%	1.00	0.78
Keyword router	83%	1.05	0.26
Every tool	95% (94.8%)	3.00	2.05

Unneeded call = a tool called that the question does not need. Total revenue needs finance data only, so every tool makes 3 calls and 2 are unneeded. Averaged over all 155 questions.

Corrections to 11/5: the 95% is calling every tool (it misses only the 8 questions that need no tool), not the keyword router plus text only; the keyword router alone is 83%.

On the current 155 questions only: the 'Both' and 'No tool' bars rest on 1 and 8 questions. Coverage rewards calling more tools, so read it next to calls per question. Gold Set v2.2: EV1R-002/006 now route to text.

Source: Week 8/results/routing_baselines.json; kpmg_eval/metrics/routing.py; ROUTING_EXAMPLES.md. Routes are Eval's labels, for Agents to review

Evaluation & Testing

Routing walkthrough | three real questions, and which tools each method calls

1 · A number from a statement

“What was Microsoft's total revenue for the three months ended March 31, 2026?”

Needs: finance data

Method	Calls	Covered	Unneeded
Text only	RAG	no	1
Keyword	Finance data	yes	0
Every tool	Finance data, RAG, GraphRAG	yes	2

Keyword router: number word 'total' → finance data

2 · A relationship (Exhibit 21)

“How many subsidiaries does Micron list in Exhibit 21, and how many are outside the US?”

Needs: GraphRAG

Method	Calls	Covered	Unneeded
Text only	RAG	no	1
Keyword	GraphRAG	yes	0
Every tool	Finance data, RAG, GraphRAG	yes	2

Keyword router: relationship word 'Exhibit 21' → GraphRAG

3 · Where the keyword router slips

“What does revenue allocated to remaining performance obligations include?”

Needs: RAG text search

Method	Calls	Covered	Unneeded
Text only	RAG	yes	0
Keyword	Finance data	no	1
Every tool	Finance data, RAG, GraphRAG	yes	2

Keyword router: number word 'revenue' → finance data

Where the keyword router sits: it is Eval's baseline for comparison, not part of the pipeline. It sends a question to GraphRAG on a relationship word (subsidiary, Exhibit 21, acquire), to finance data on a number word (revenue, total, margin), to RAG on an explanation word (why, describe). The Agents' orchestrator does the real routing, and they can score it on the same 155 questions with our script (next line).

For Agents: python scripts/run_routing_baselines.py --router name=your_module.py:route, or --calls name=calls.jsonl from your logs. Seven worked examples in ROUTING_EXAMPLES.md.

Source: Week 8/results/routing_examples.json (scripts/routing_examples.py); scripts/run_routing_baselines.py

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Typos and messy wording | 427 rewrites of the 155 questions, same meaning, scored like the clean ones

Rewrite (n)	Example	Scope check, no company picked	Same, company picked in the UI	Router: needed tools	RAG search	GraphRAG search
Company misspelled (129)	Microsfot, Mircon, Net App	**98% → 19%**	98% → 93%	84% → 80%	93% → 93%	37% → 37%
Finance word misspelled (143)	revnue, opertaing, subsidaries	82% → 82%	99% → 90%	83% → 77%	91% → 80%	30% → 17%
Casual: ticker, lower case (155)	what was msft total revenue for ...	**82% → 9%**	97% → 90%	83% → 76%	92% → 91%	32% → 32%

94%

right decisions on misspelled names once the scope check also matches tickers and near-miss spellings (79% on casual); no change on the clean questions

What drops, and why

Company names: the check knows exact names only, so 'Microsfot' or 'msft' turns an answer into 'which company?'.

Finance words: RAG's search loses 8 and GraphRAG's 9 of 70 questions; keyword matching does not forgive 'revnue'.

Searches ignore name typos because the company comes in as a filter.

For Agents: match tickers and near-miss names before routing, and correct spelling before search. Scores are clean → messy, on the current 155 questions; searches on the Microsoft 10-Q questions only.

Source: Week 8/results/messy_wording.json, data/messy_wording_v1.jsonl (scripts/run_messy_wording.py). Scope check and router are Eval's reference rules, not the orchestrator

Evaluation & Testing

Next steps | after round 1: one ask per team

Team	Ask (from this week's results)
RAG	Run our narrative judge with your key (or lend one); mark answers that state the figure as answered; try candidate k and fusion weights on the 6 near misses
GraphRAG	Rank by period and break ties (#68: right number in the top 10 on 63/93, at rank 1 on 26); re-run the graph arm on v2.2 with #70; check 6 rows of the hops sheet; Exhibit 21 into Aura
Agents	Always send ticker and accession; split cross-company and 10-K + 10-Q questions into one search each; match tickers and near-miss names; run your orchestrator through our scorer
Data	Round 1 ran on 20 filings (NetApp's 10-Q held for review); now load Microsoft's FY2024 and FY2025 10-Ks, already in your inventory (joint ask with GraphRAG)
Eval	Hand-label the 28 narrative answers to validate the judge; score the Agents' first end-to-end run with the same grader; add the new candidates after round 1; grow the 1 'both' and 8 'no tool' routes

Thanks: RAG ran round 1 twice and GraphRAG ran both arms, and both traced their misses, which is how we found our errors. RAG's retriever and GraphRAG's fact loader ran here unchanged, and GraphRAG's Exhibit 21 output fed the relationship check.

Source: Week 8/FOR_TEAMS_W8.md (one section per team, with file paths)